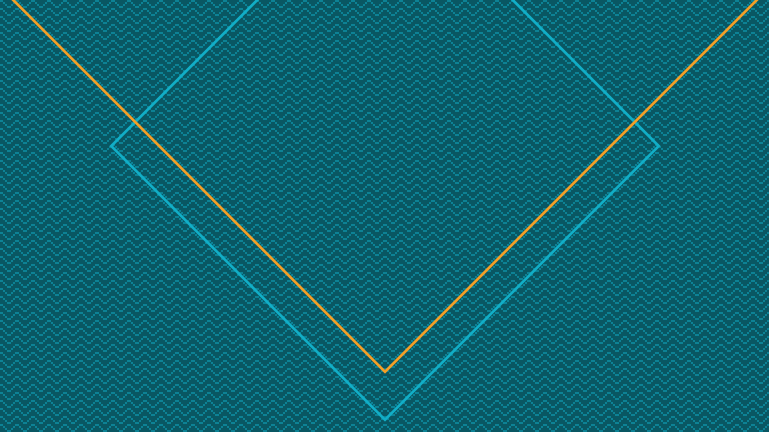


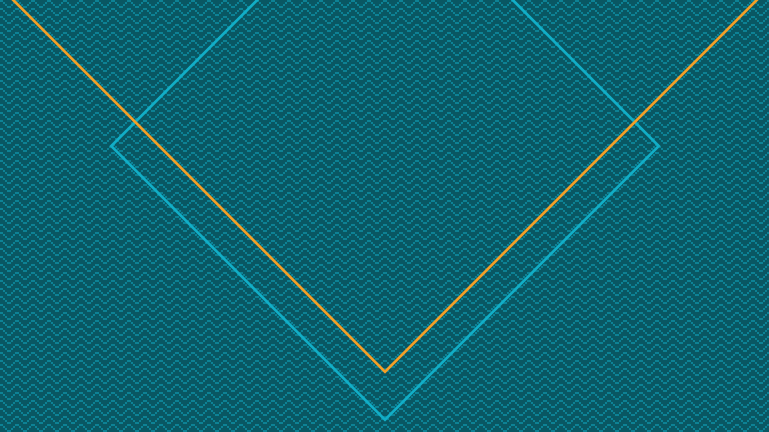


QuTe FDS

Quantum Teleportation induced
Frequency Dependent Squeezing

Candidato
Andrea Grimaldi

Sede Ospitante
INFN Padova

A decorative graphic at the top center of the slide, consisting of two overlapping diamond shapes. The outer diamond is a light blue color, and the inner diamond is a slightly darker blue. They are centered and point downwards.

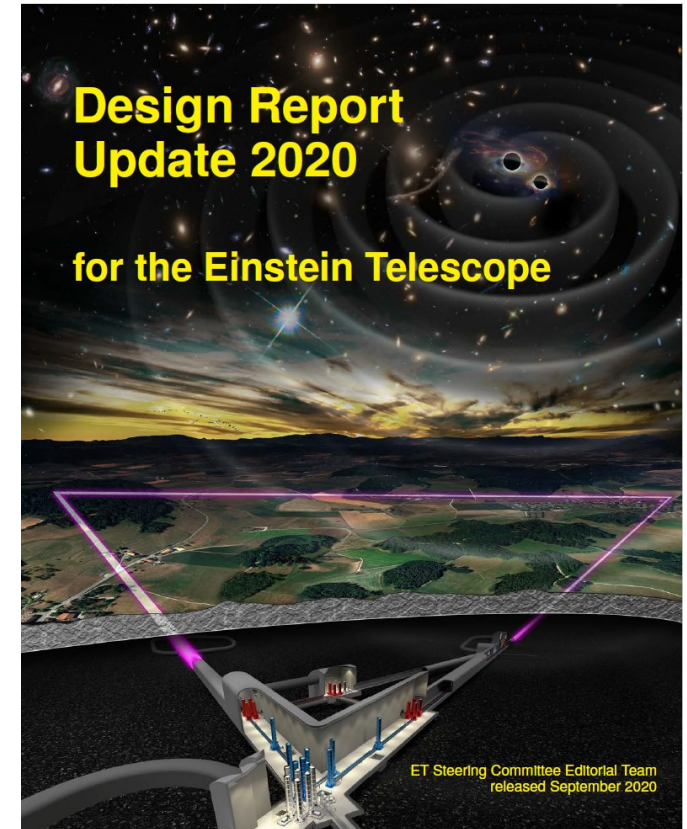
WHY?

Why For the Science

The new Generation of Gravitational Wave Detectors is extending the Sensitivity to Low Frequency (8~20Hz).

Improving different aspect of the interferometer:

- Suspension, Coating, Vacuum, ...
- New Detector Configuration: **Detuned interferometer**



Heavier
systems

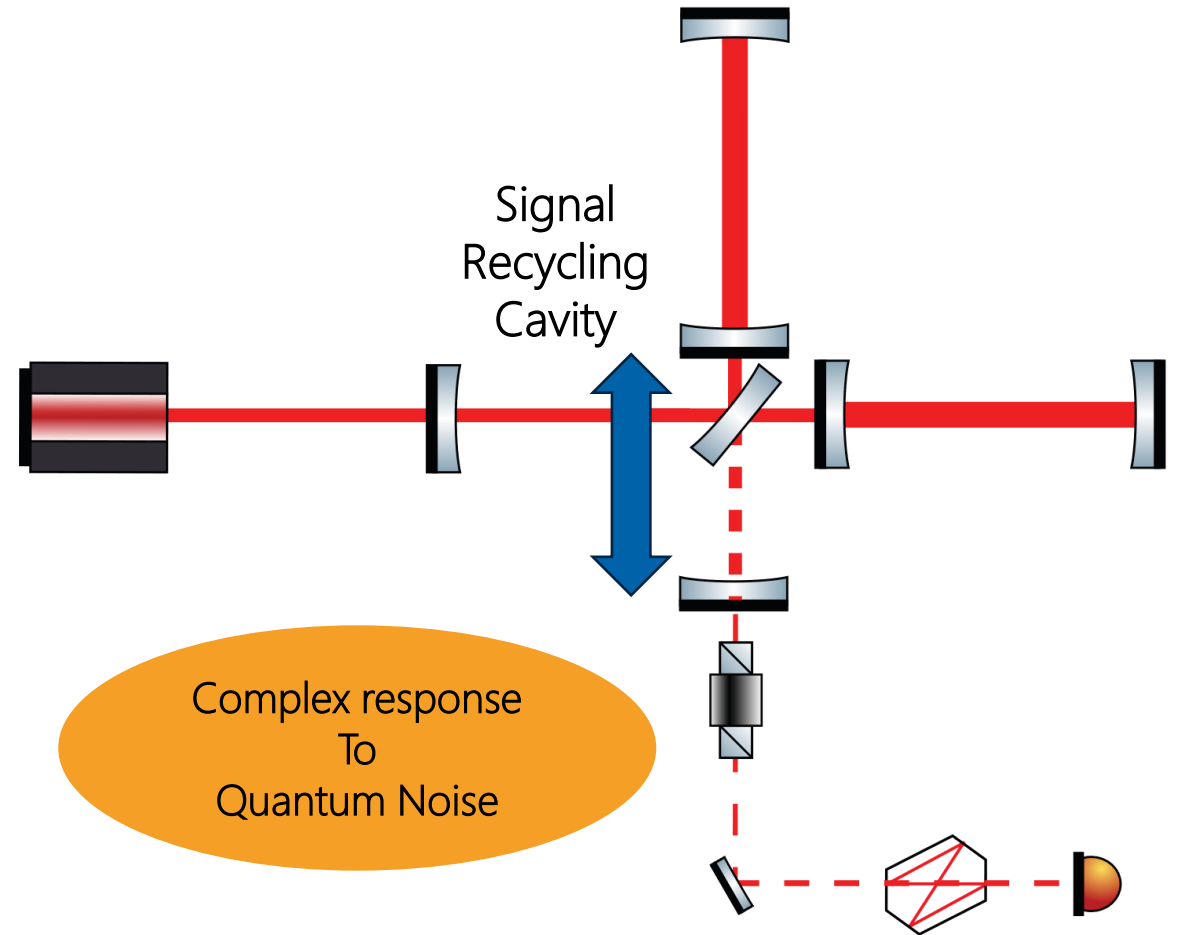
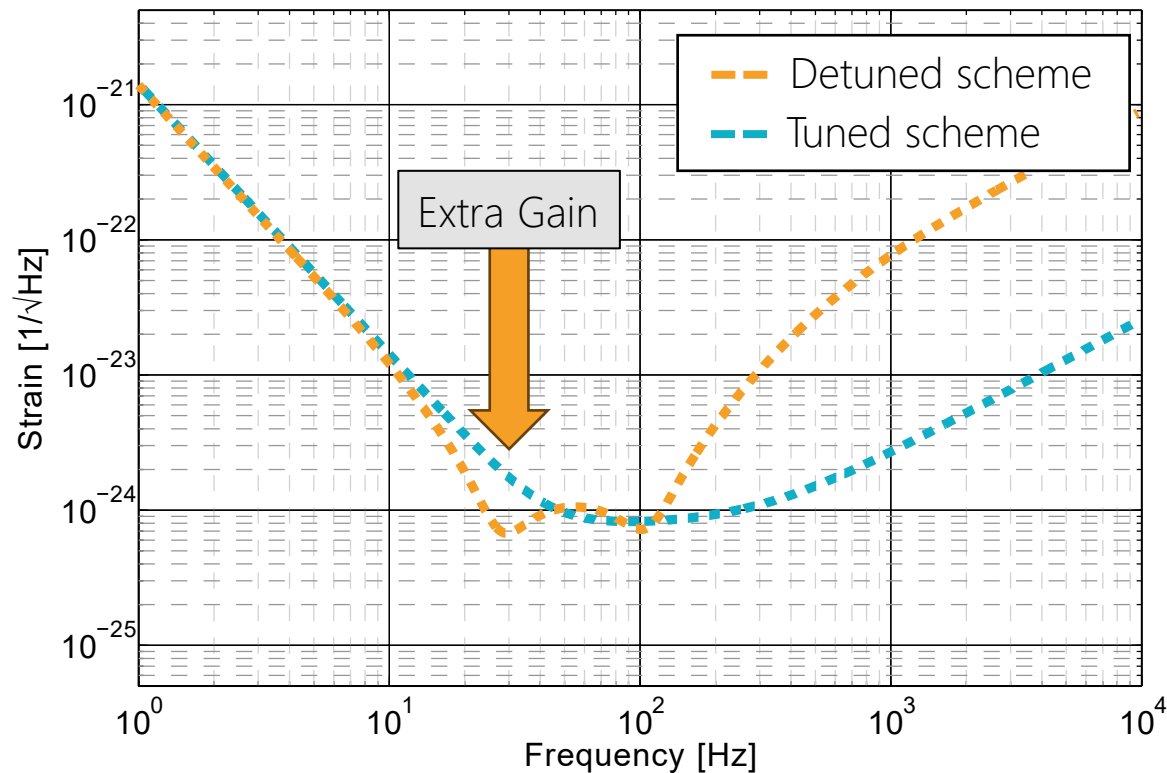
Warning for
electromagnetic/neutrino
follow-up

Core collapse
supernovae

Why For the Science

Detuned Configuration

The Signal Recycling Cavity is detuned to enhance **Low-Frequency Response**



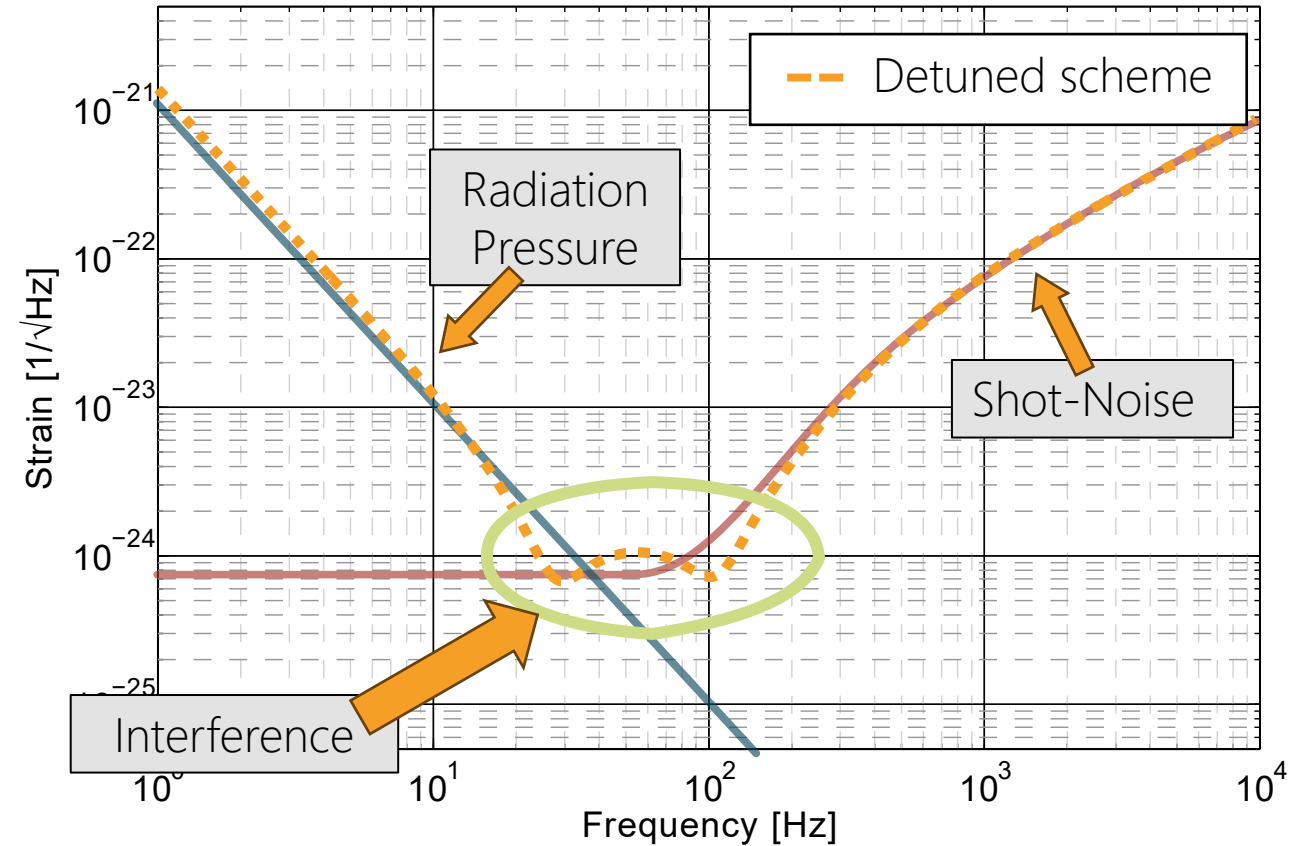
Why For the Science

Quantum Noise

Vacuum Entering from the output

- Radiation Pressure noise ($1/f^2$) proportional to Amplitude Quadrature
- Shot-Noise ("flat") proportional to Phase Quadrature

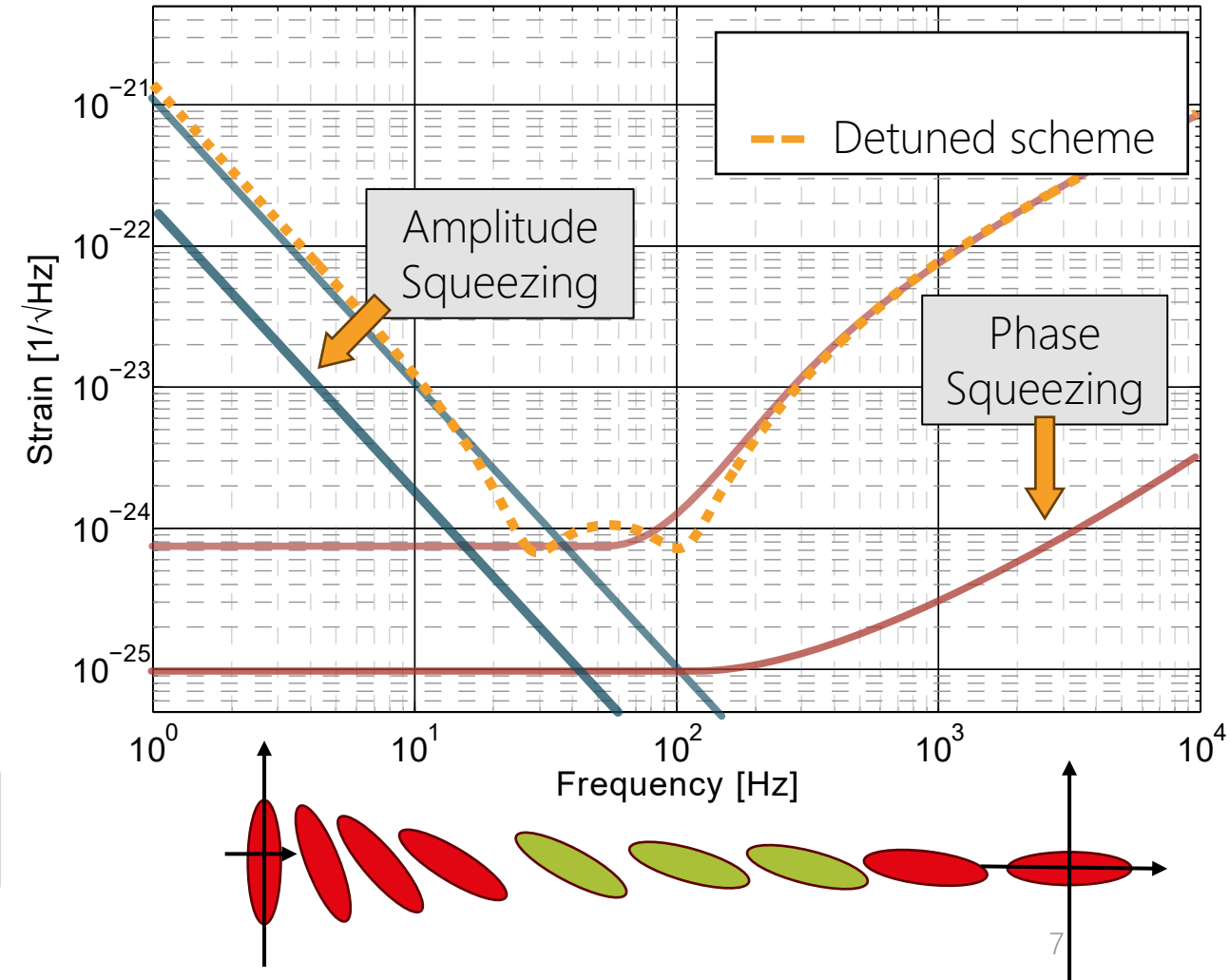
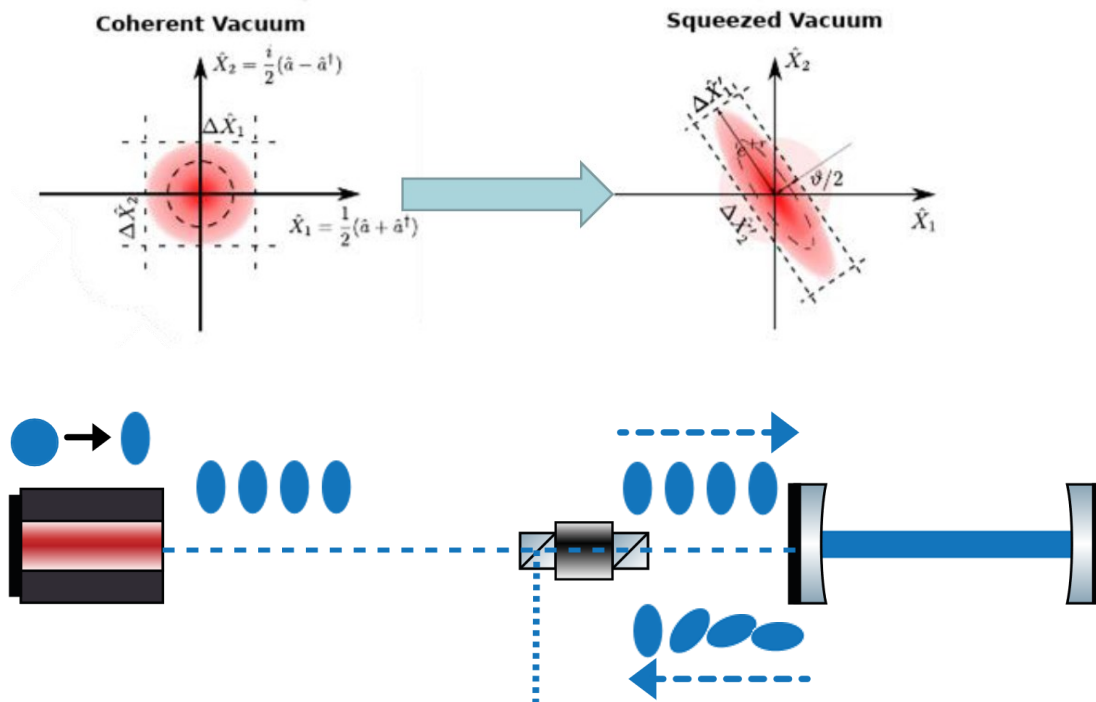
In a detuned interferometer **Noise are correlated**



Why For the Science

Squeezed Vacuum Injection

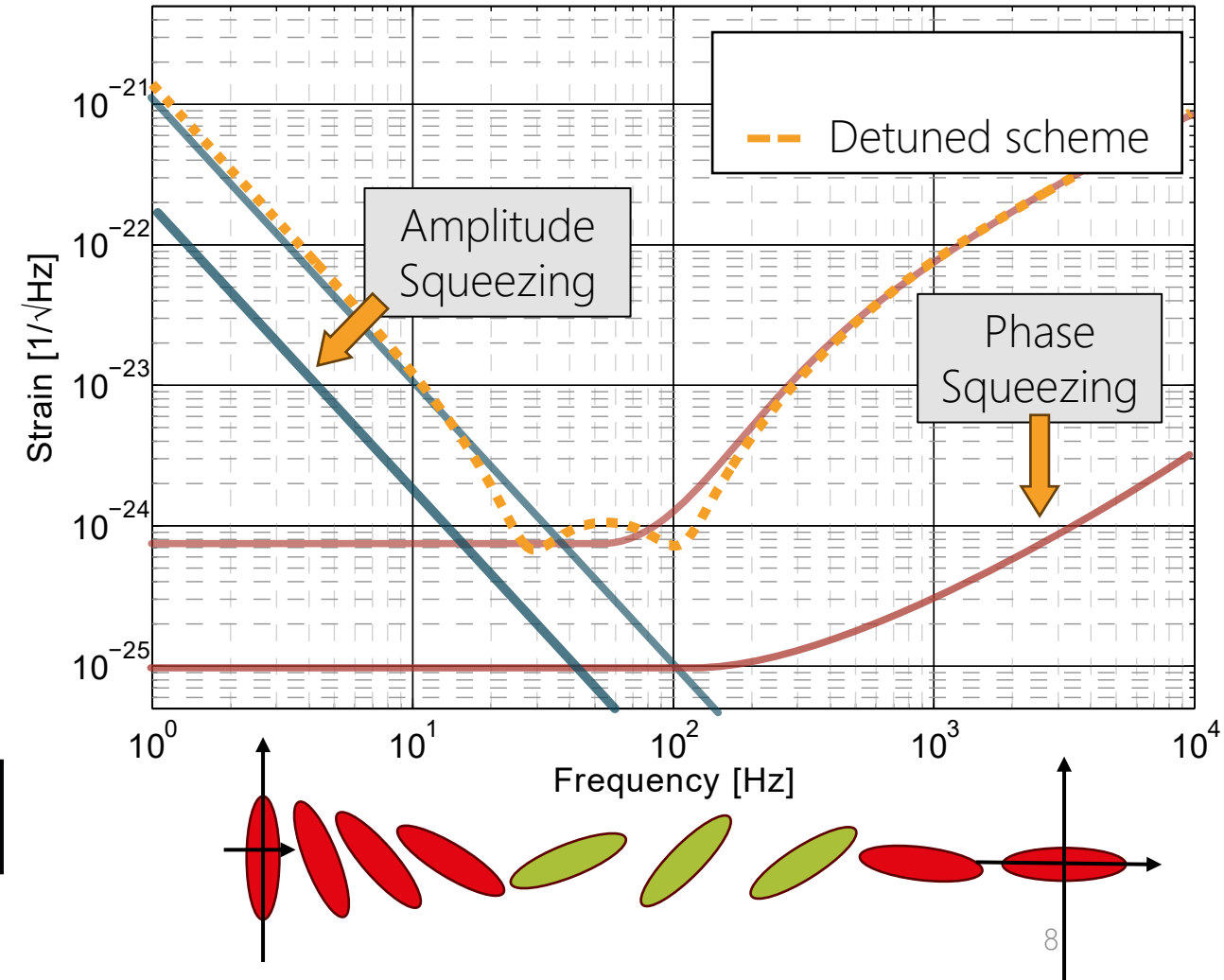
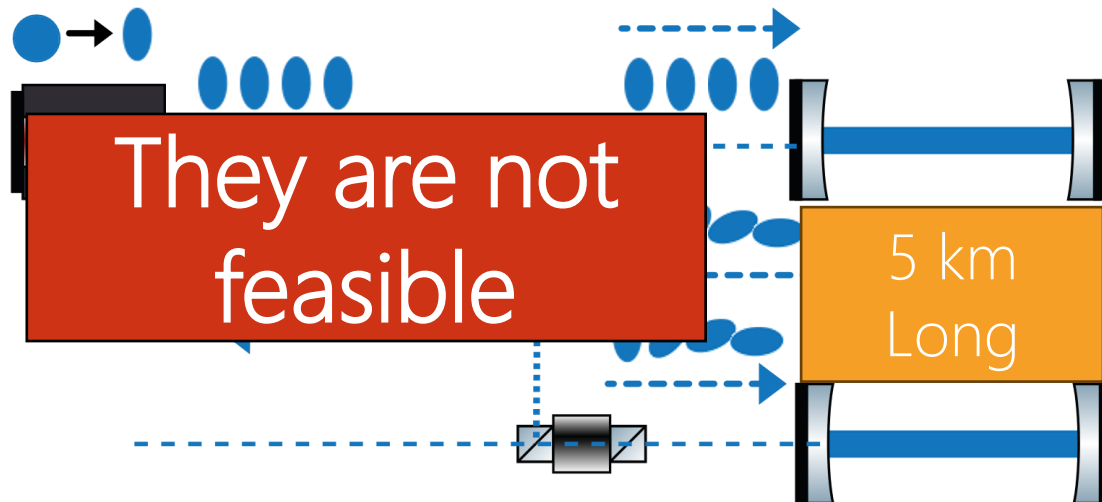
Replace the vacuum with
Frequency Dependent Squeezed
Vacuum

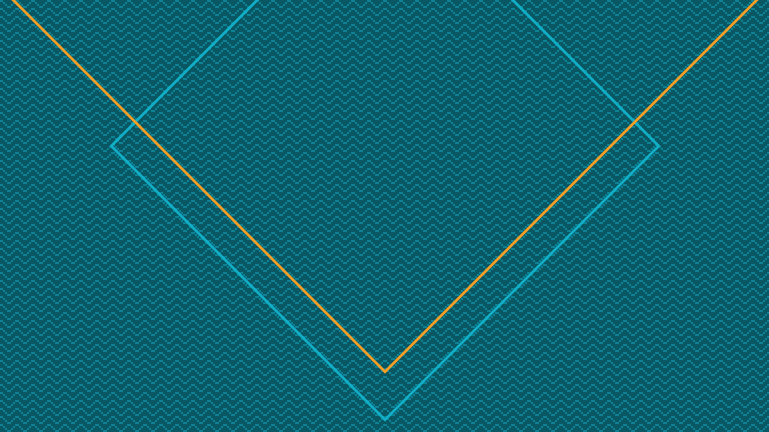


Why For the Science

Squeezed Vacuum Injection

Second rotation to compensate
for the Detuned Signal Recycling

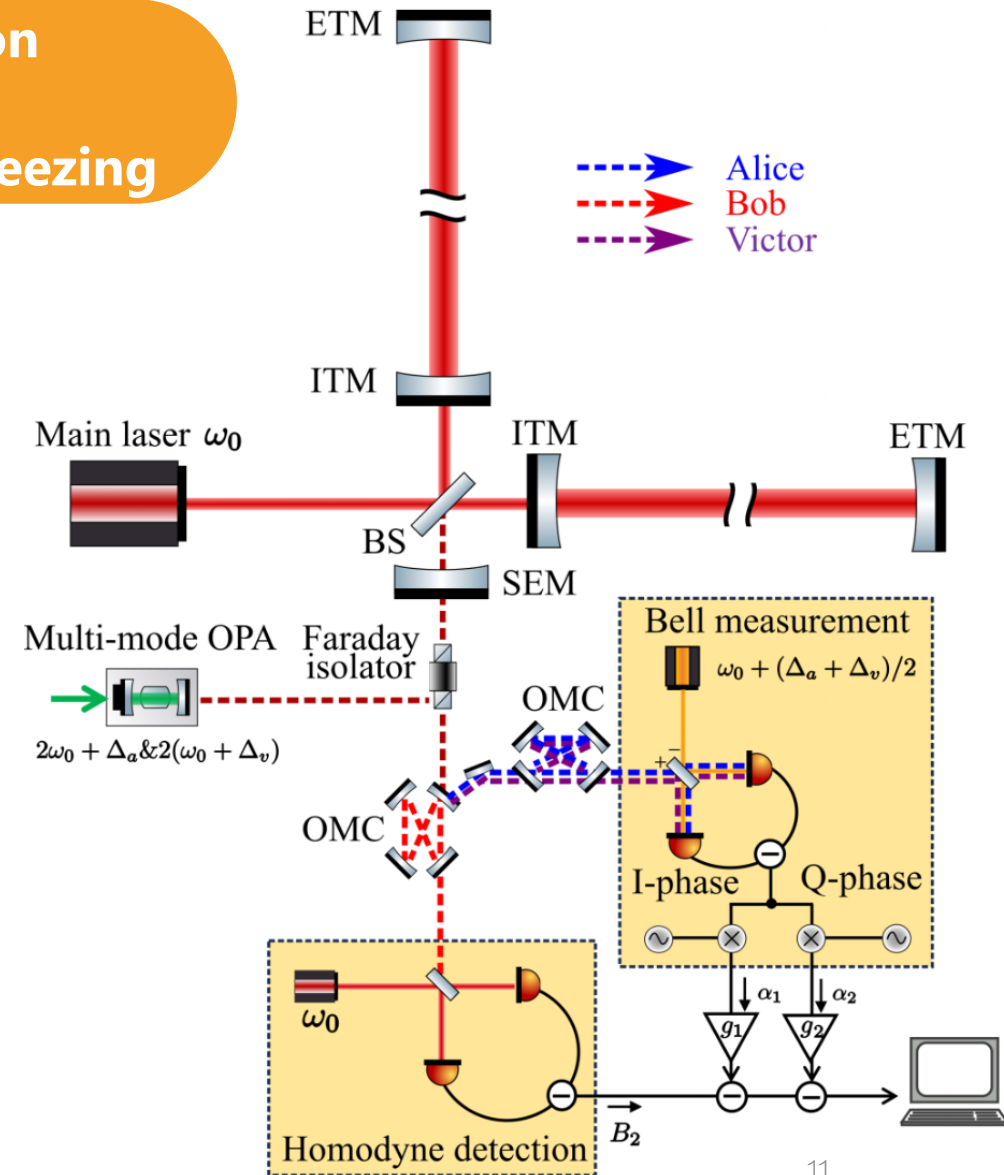
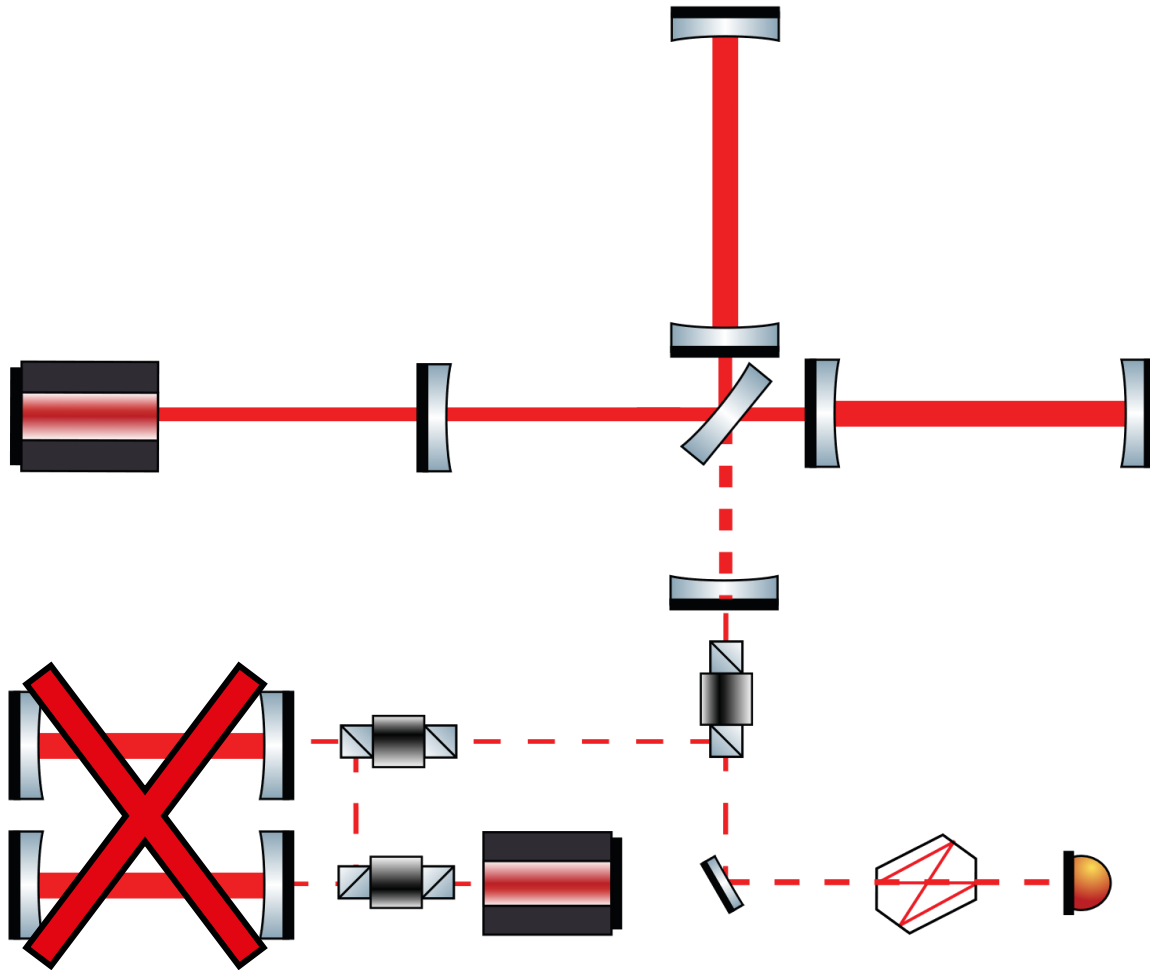


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WHAT?

What Qute_FDS

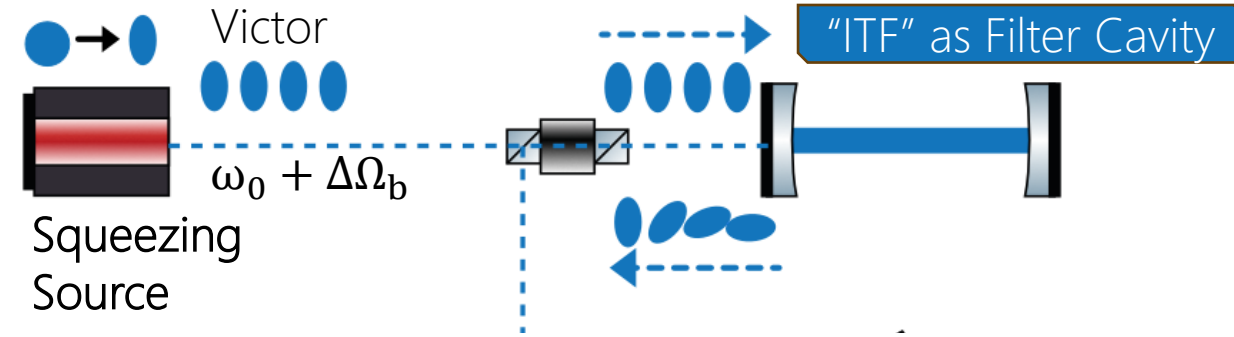
Quantum Teleportation induced Frequency Dependent Squeezing



What Qute_FDS

Steps

1. Generate **Frequency Dependent Squeezing** at detuned frequency $\omega_0 + \Delta\Omega_b$ by using the **Interferometer as Filter Cavity**.
2. Couple it with **EPR entangled** link and project the joint state with the Bell Measurement
3. **Teleport** the Frequency Dependent Squeezing from the $\omega_0 + \Delta\Omega_b$ to the interferometer frequency ω_0
4. Inject the teleported state into the **Interferometer ITF**



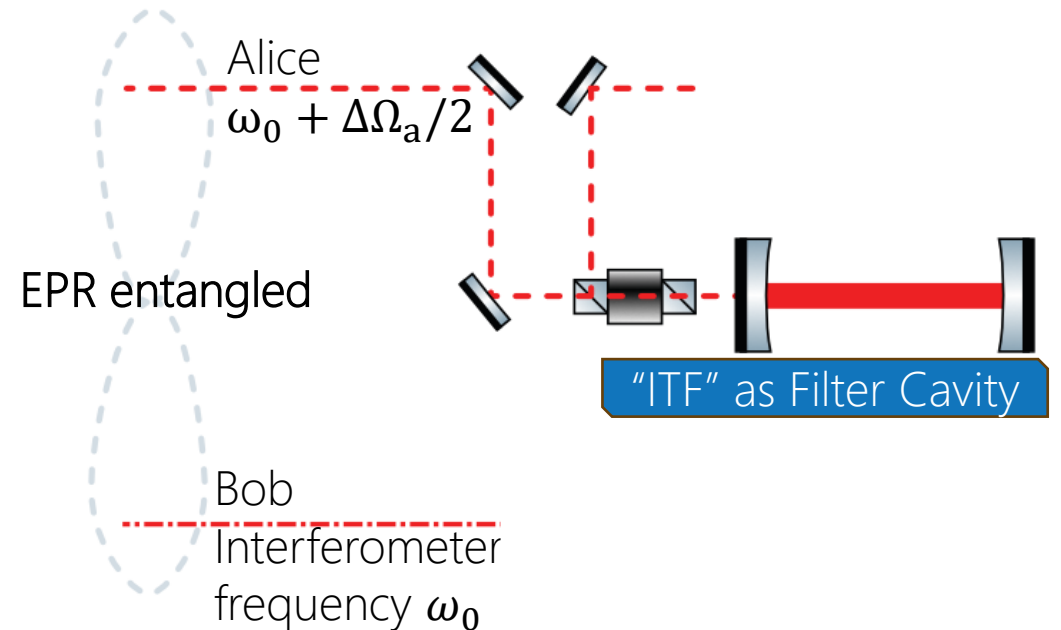
The Entangled Link

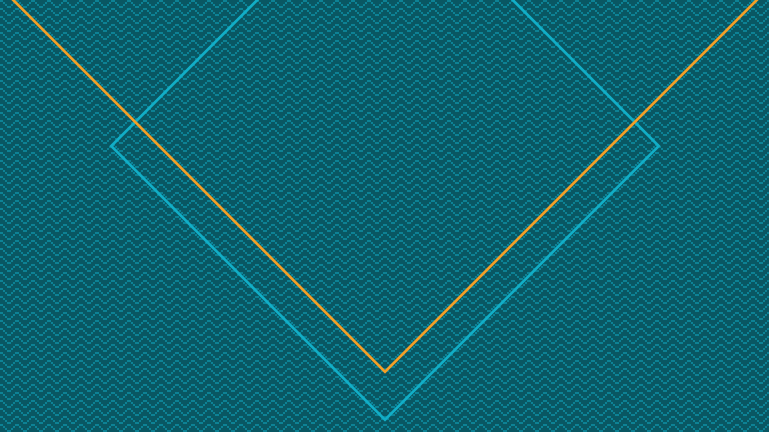
Prepare a Frequency Dependent EPR entangled link by using the Interferometer as Filter Cavity

1. Alice generated at $\omega_0 + \Delta\Omega_a/2$
2. Bob generated at ω_0

Alice interacts with the Interferometer at detuned frequency $\omega_0 + \Delta\Omega_a/2$.

Second Rotation Imprinted in the EPR Link



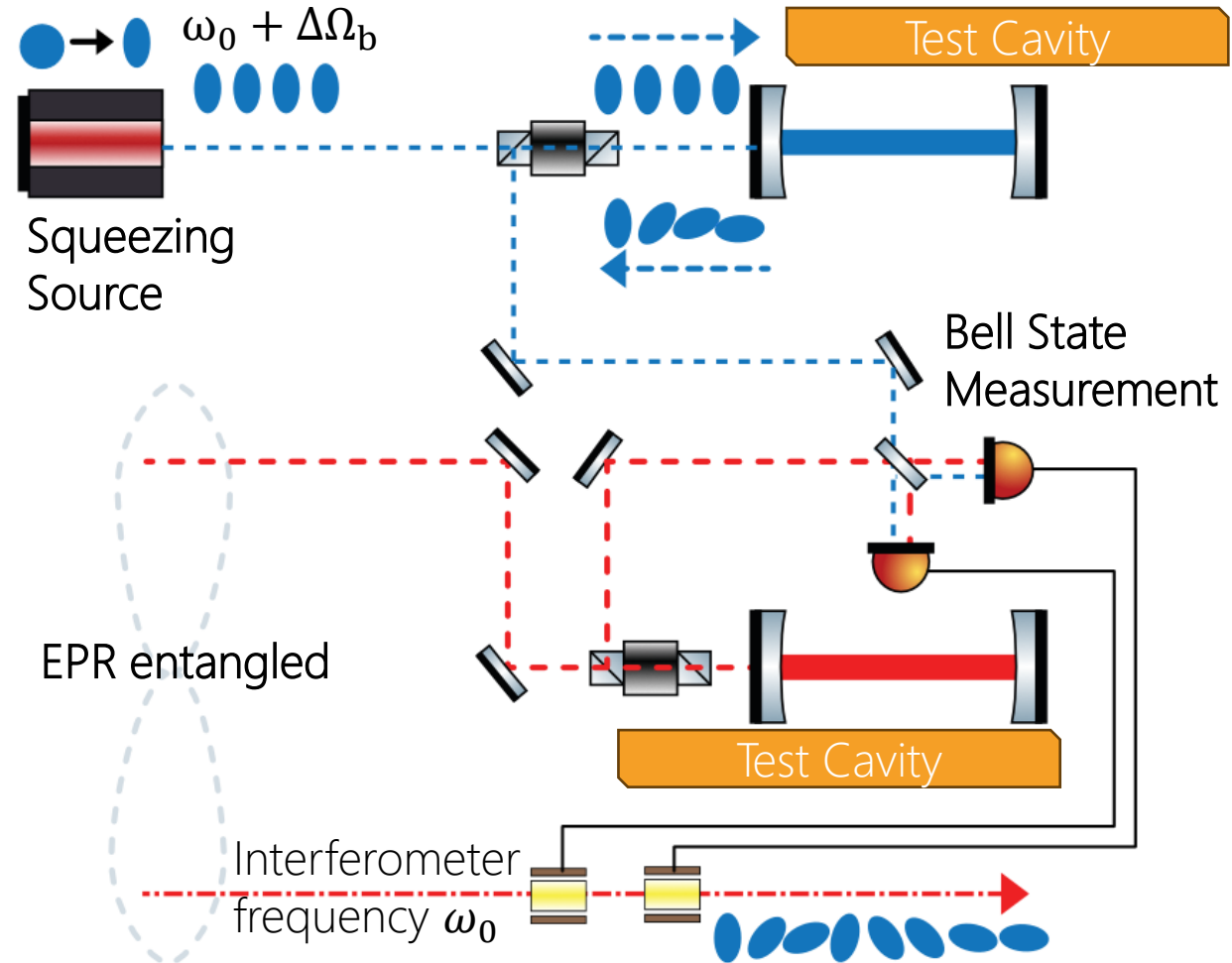
A decorative graphic at the top center of the slide, consisting of two overlapping diamond shapes. The outer diamond is a light blue color, and the inner diamond is a slightly darker blue. They are oriented with their vertices pointing towards the top and bottom.

HOW?

What Qute_FDS

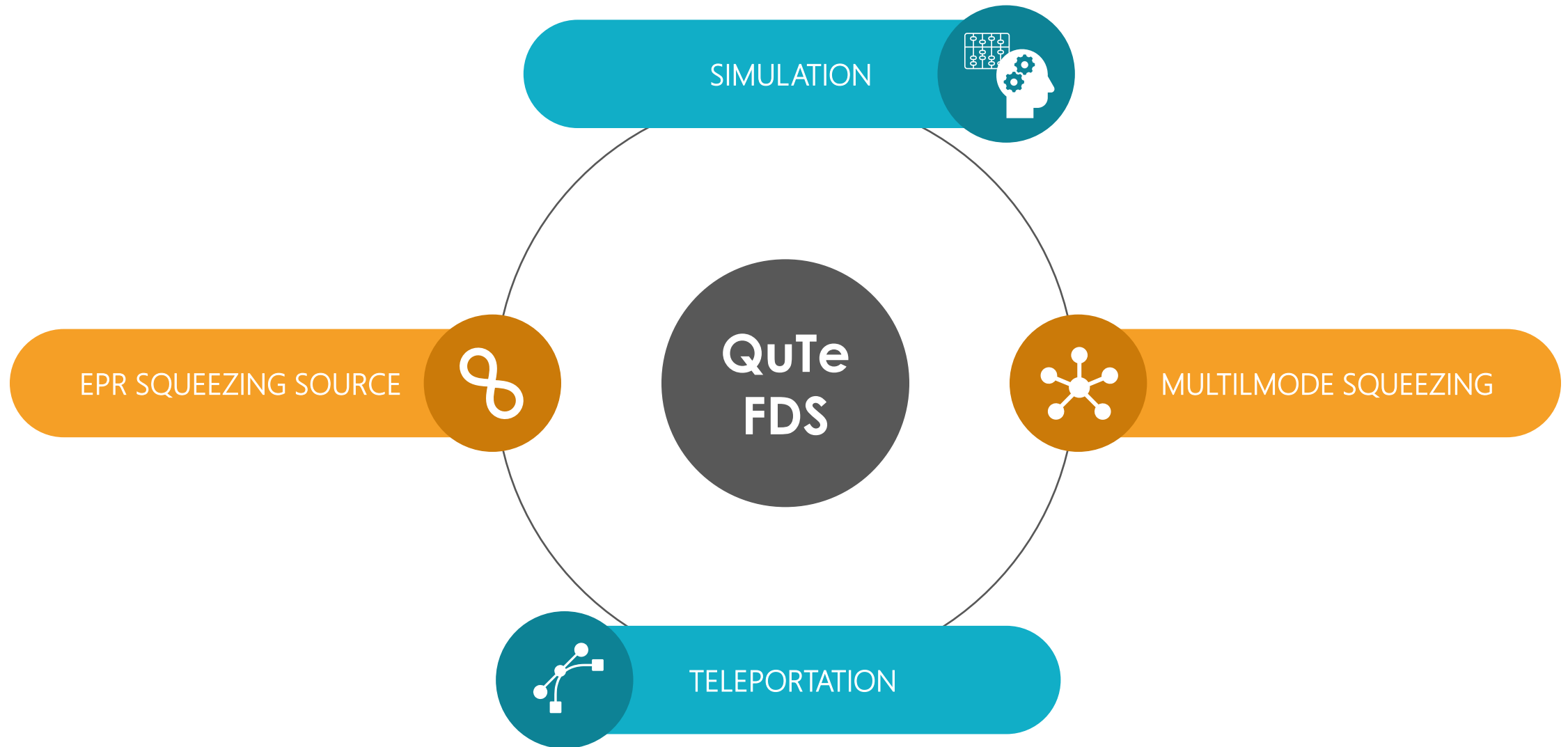
Table-Top Experiment

1. Interferometer replaced by a Linear Cavity
2. Phase Rotations around 1MHz
3. Double Rotation measured by Tomography of the final state



HOW?

Working Packages



Gantt



MODEL
 Simulations of the
 protocol
 Re-adaptation
 with a linear cavity

**TECHNICAL
 ANALYSIS**
 Estimation
 parameters

 Milestones

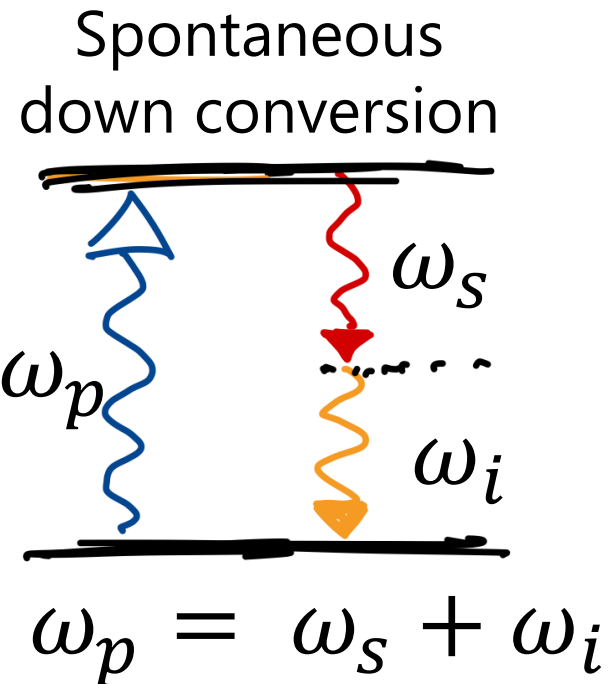
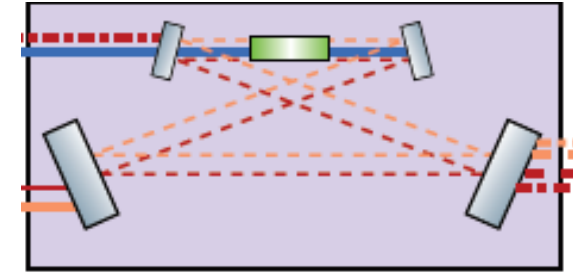
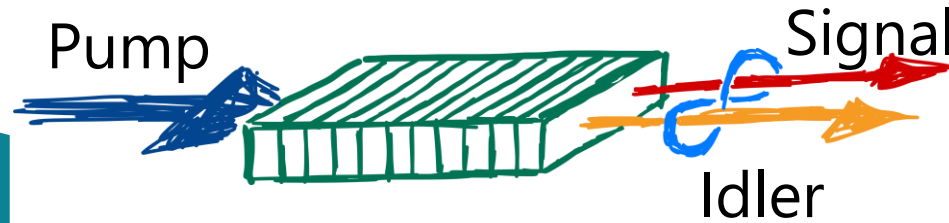
M1 Final design of the optical setup and place the order for the custom optics.

HOW?

EPR source installation



Installation
of the EPR
source



Optical Resonator select ω_s and ω_i :

$\omega_s \neq \omega_i$ we generate a entangle pair

$$\omega_p = 2 \omega_0 + \Delta\Omega_a$$

$$\omega_s = \omega_0$$

$$\omega_i = \omega_0 + \Delta\Omega_a$$

HOW?

EPR source installation



Installation
of the EPR
source

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Milestones

M1 Final design of the optical setup and place the order for the custom optics.

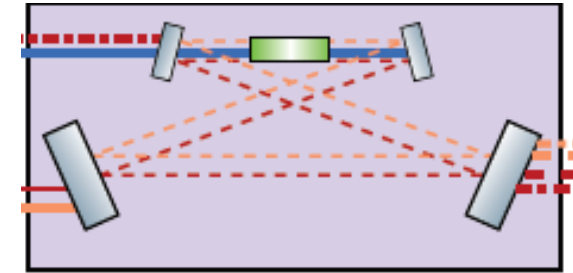
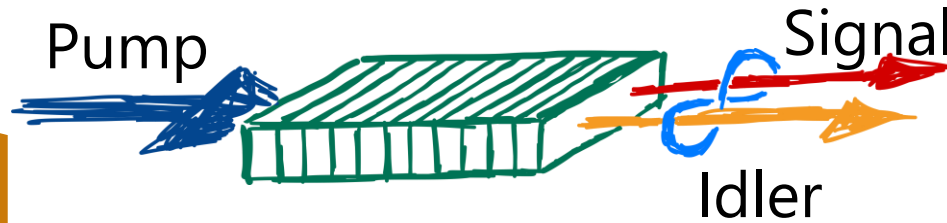
M2 Measurement of two-mode squeezing

HOW?

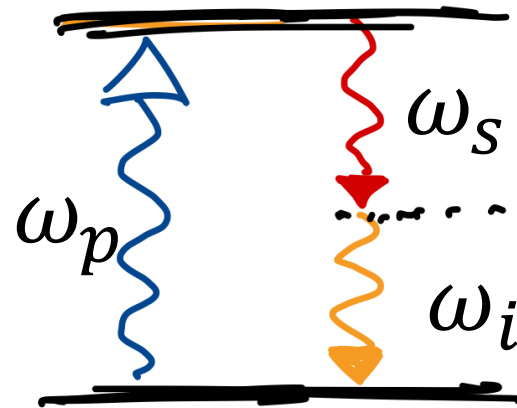
Multi-mode Squeezing source



Modification to
a Single Mode
Squeezer



Spontaneous
down conversion



$$\omega_p = \omega_s + \omega_i$$

Optical Resonator select ω_s and ω_i :

$\omega_s = \omega_i$ we generate a Squeezing

$$\omega_p = 2 (\omega_0 + \Delta\Omega_b)$$

$$\omega_s = \omega_i = \omega_0 + \Delta\Omega_b$$

HOW?

Multi-mode Squeezing source



Modification to
a Single Mode
Squeezer

HOW?

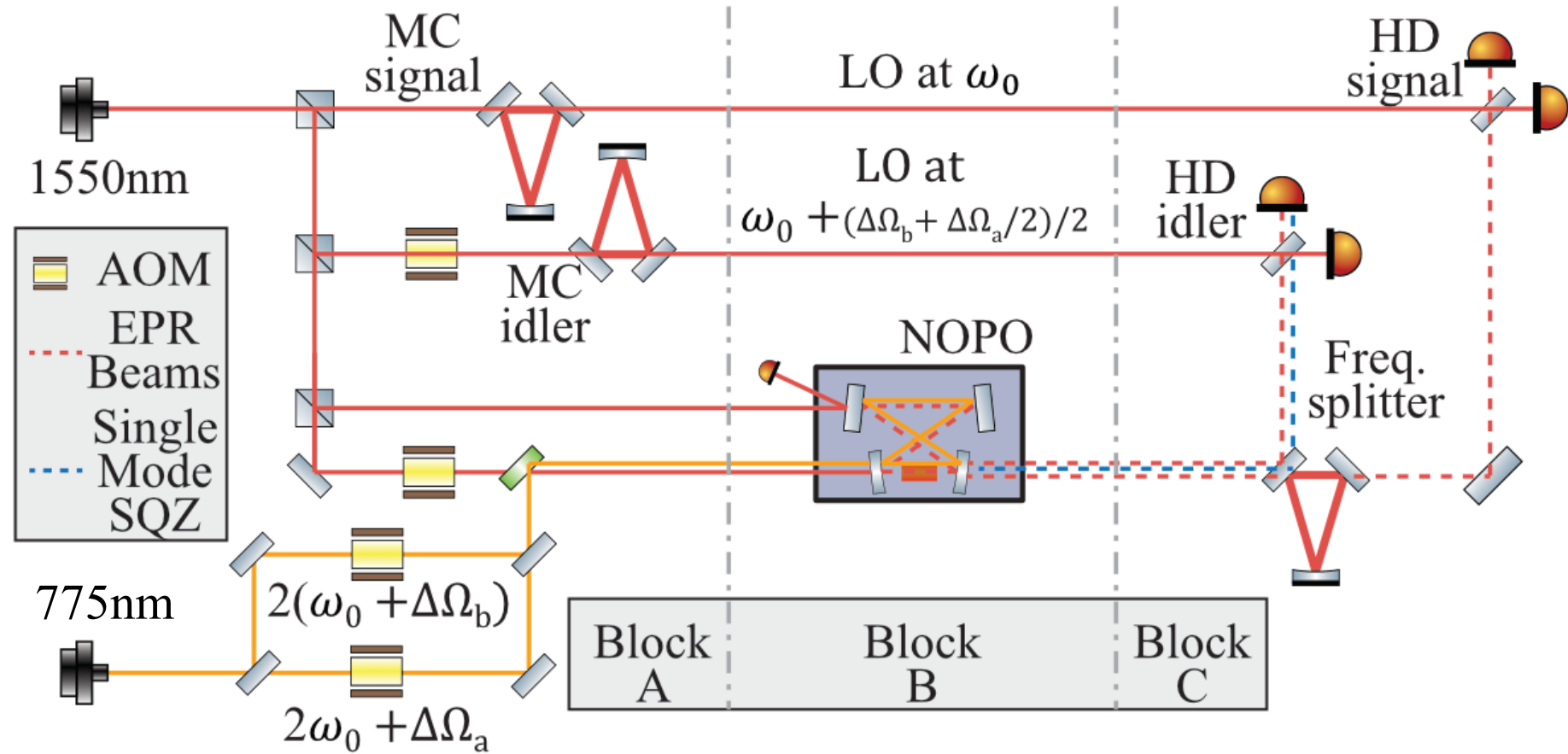
Multi-mode Squeezing source



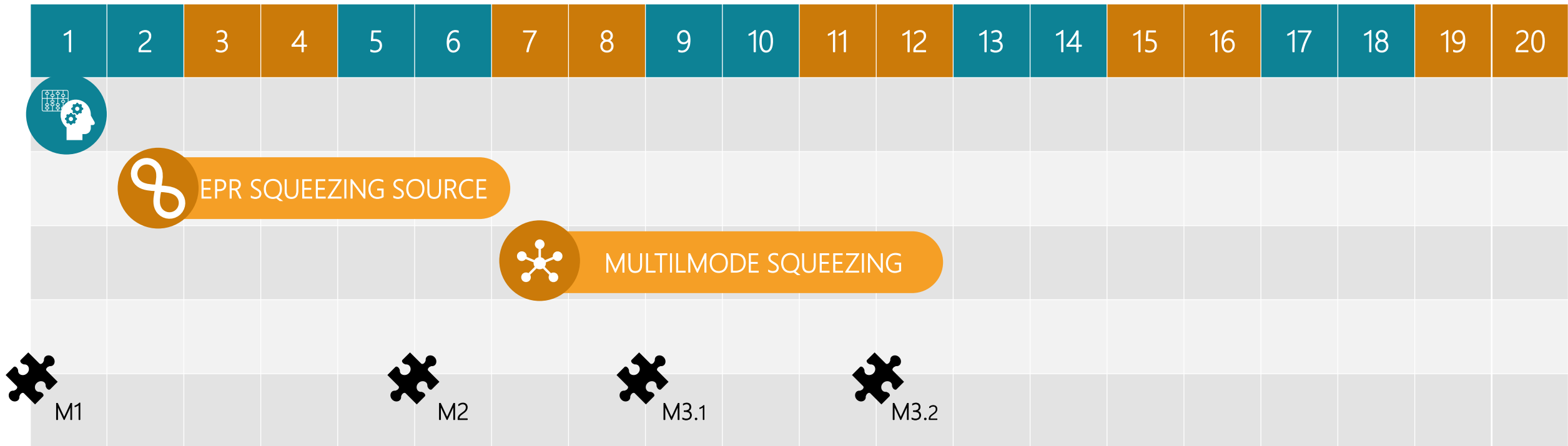
Modification to
a Single Mode
Squeezer



Test Multimode
Mode Squeezer



Gantt



Milestones

M1 Final design of the optical setup and place the order for the custom optics.

M2 Measurement of two-mode squeezing

M3.1 Measurement of single mode squeezing

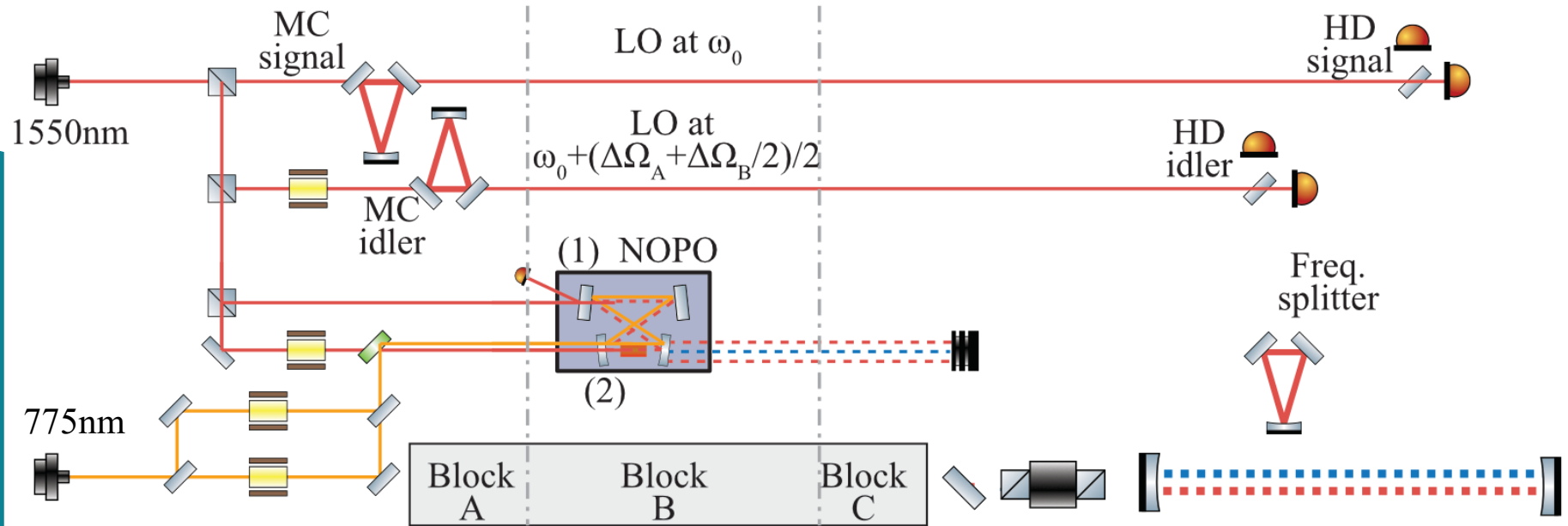
M3.2 Measurement of the benchmark defined in M2 and M3.1 simultaneously.

HOW?

Multi-mode Squeezing source

Validation of the
protocol

In three
"simple"
steps

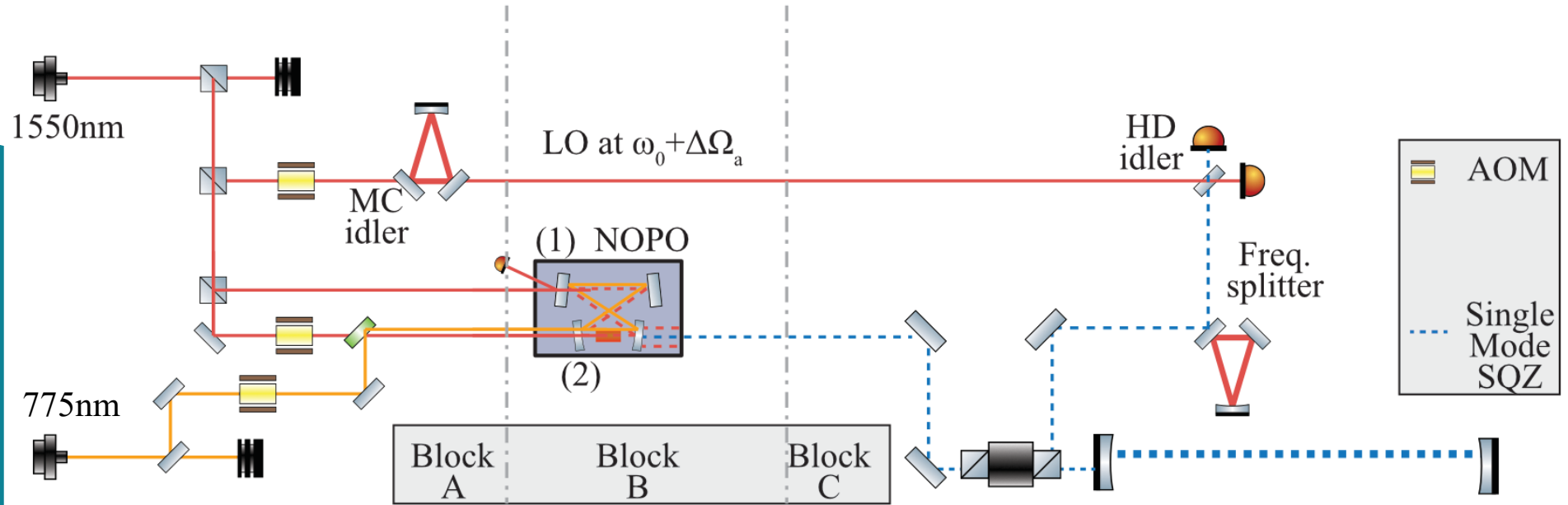


HOW?

Multi-mode Squeezing source

Validation of the
protocol

In three
"simple"
steps



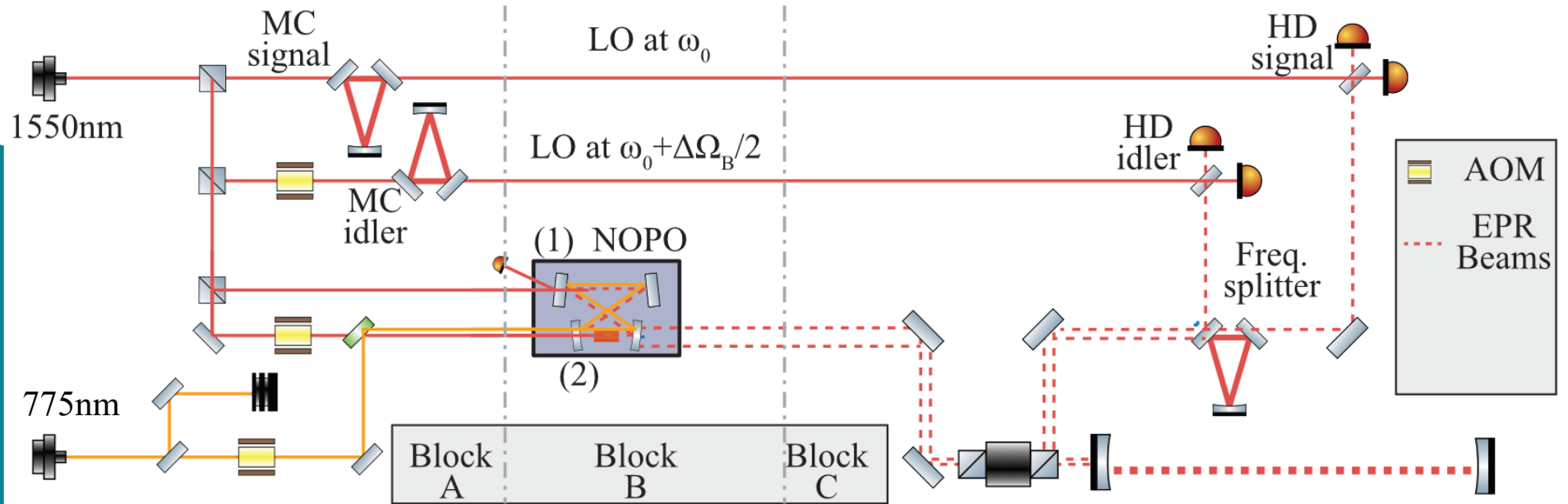
Single Mode
Frequency
Dependent Squeezing

HOW?

Multi-mode Squeezing source

Validation of the
protocol

In three
"simple"
steps



Single Mode
Frequency
Dependent Squeezing

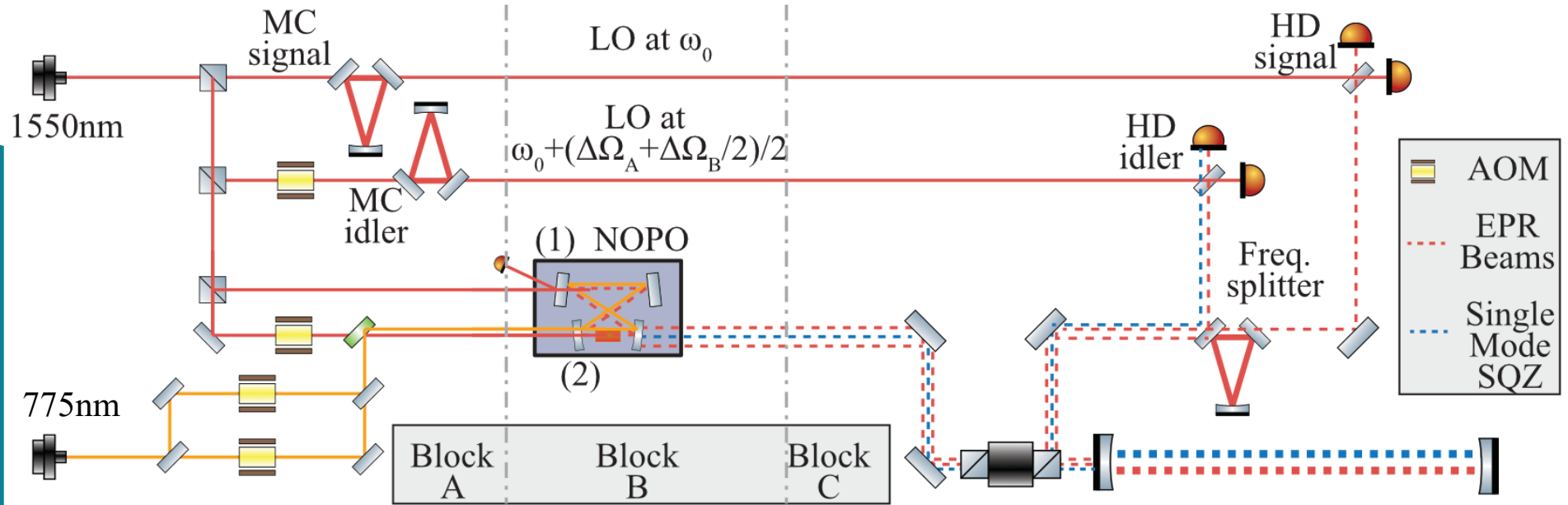
EPR Conditional
Frequency
Dependent Squeezing

HOW?

Multi-mode Squeezing source

Validation of the protocol

In three "simple" steps

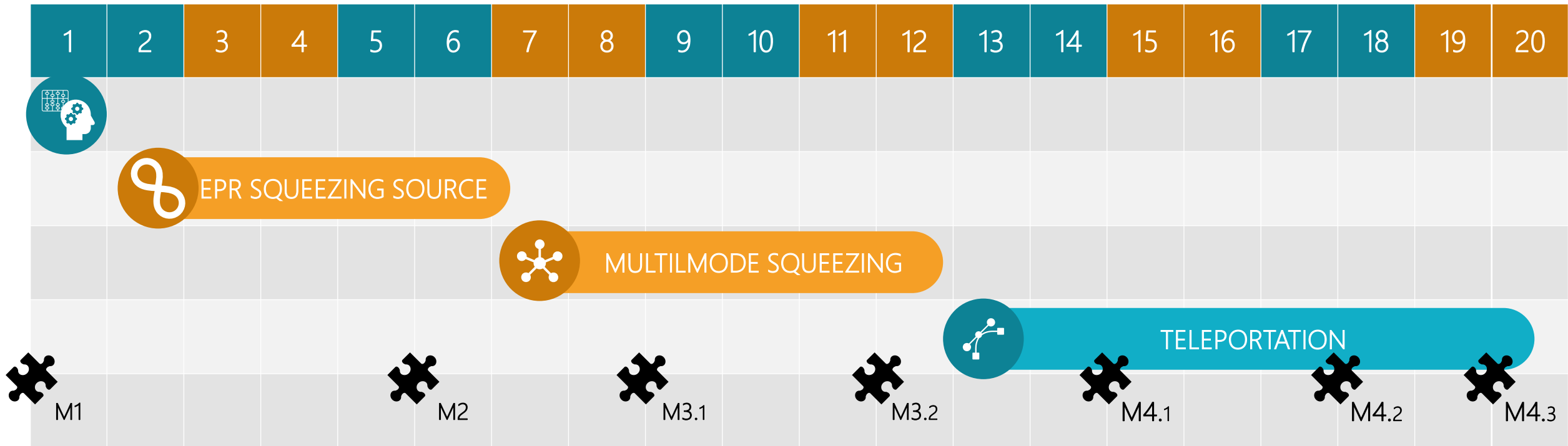


Single Mode
Frequency
Dependent Squeezing

EPR Conditional
Frequency
Dependent Squeezing

Quantum
Teleportation induced
Frequency
Dependent Squeezing

Gantt



Milestones

M1 Final design of the optical setup and place the order for the custom optics.

M2 Measurement of two-mode squeezing

M3.1 Measurement of single mode squeezing

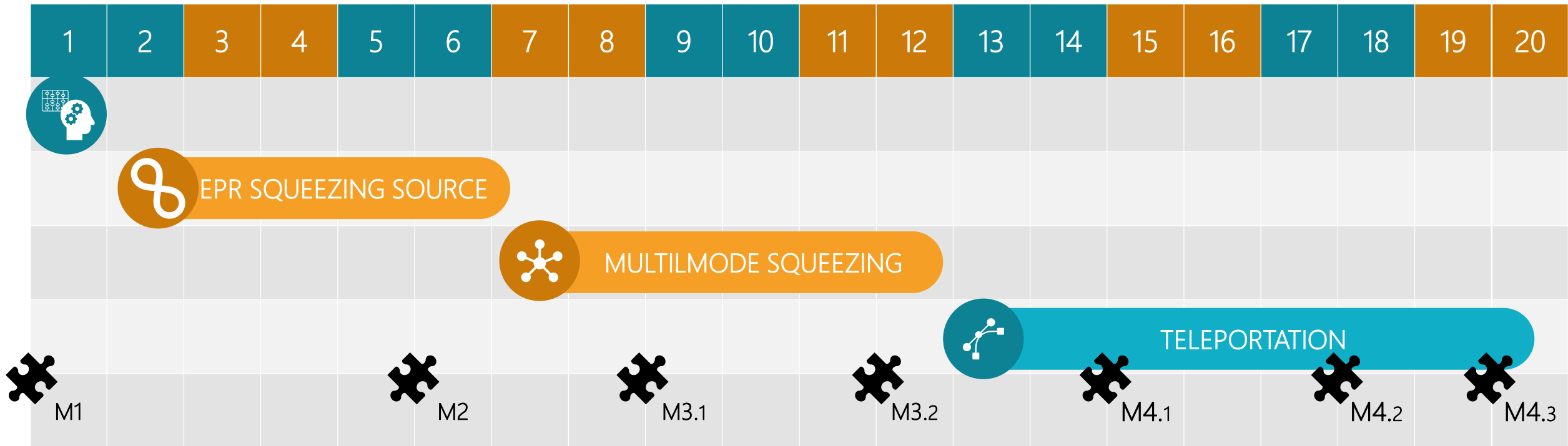
M3.2 Measurement of the benchmark defined in M2 and M3.1 simultaneously.

M4.1 Measurement of Frequency-Dependent Squeezing

M4.2 Measurement of Conditional Frequency-Dependent Squeezing

M4.3 Measurement of Quantum Teleportation induced Frequency-Dependent Squeezing

Gantt



Plan Defined on 20 months instead of 24.

Risk Management & Contingency Plans:

CP1 Piezo actuators delivery time

CP2 Alternative scheme for the multimode squeezer

Contingency time: 4 months

Milestones

M1 Final design of the optical setup and place the order for the custom optics.

M2 Measurement of two-mode squeezing

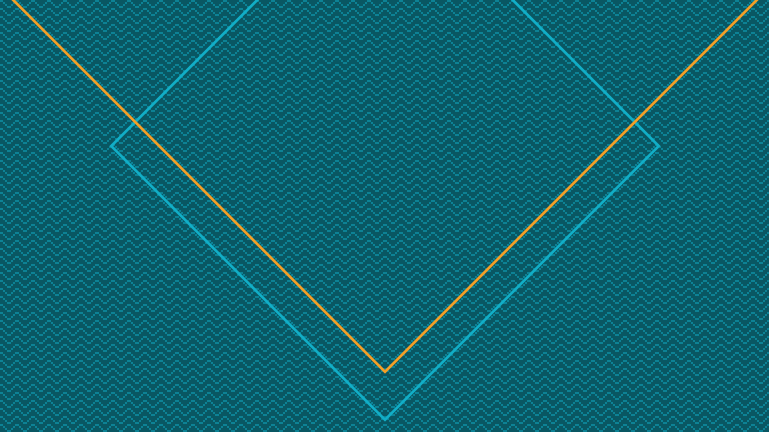
M3.1 Measurement of single mode squeezing

M3.2 Measurement of the benchmark defined in M2 and M3.1 simultaneously.

M4.1 Measurement of Frequency-Dependent Squeezing

M4.2 Measurement of Conditional Frequency-Dependent Squeezing

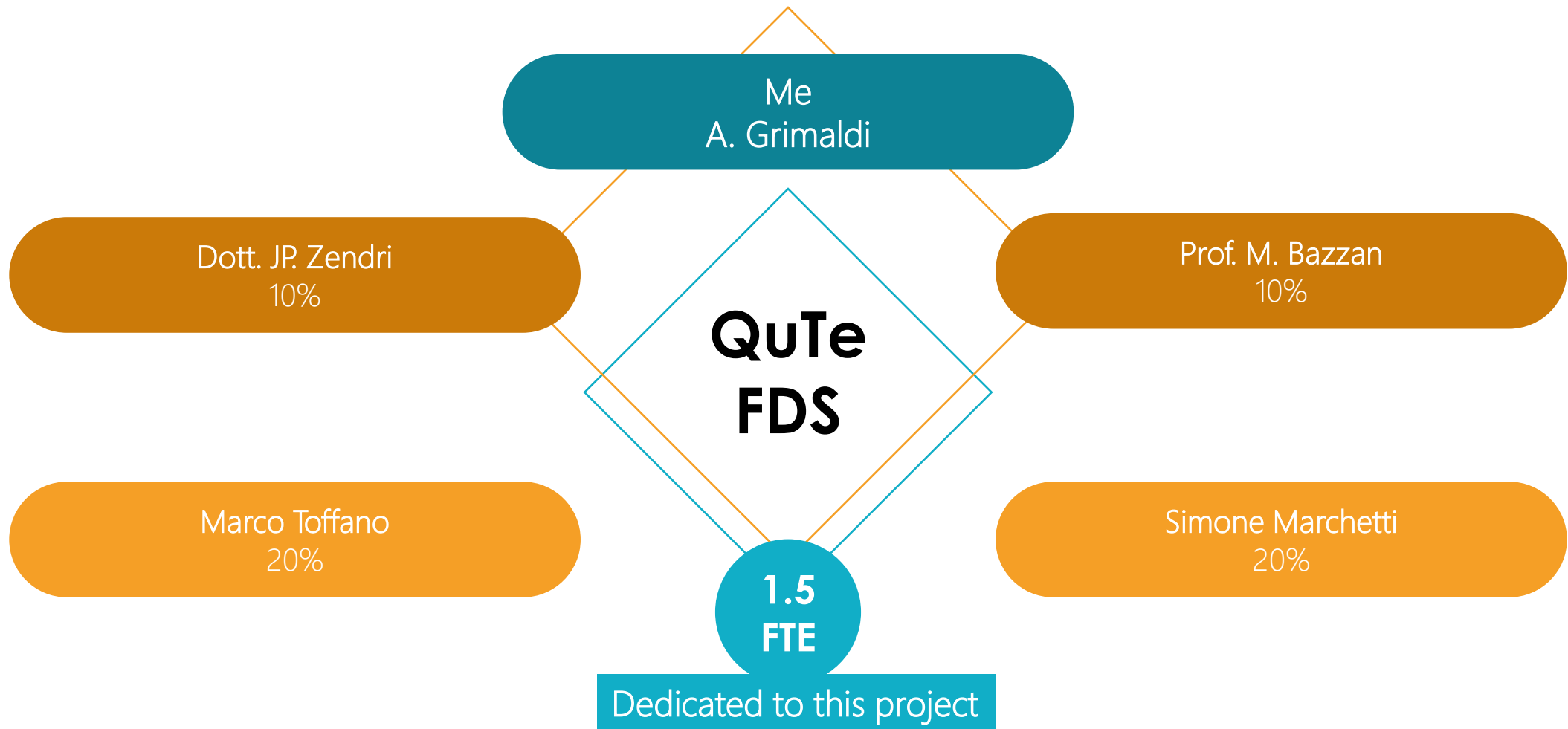
M4.3 Measurement of Quantum Teleportation induced Frequency-Dependent Squeezing

A decorative graphic consisting of several overlapping, thin, light blue and orange lines forming a series of nested, downward-pointing chevrons or a stylized 'V' shape, centered at the top of the slide.

WHO?

WHO?

Official Team



Synergy with the individual projects

WHO?

Official Team

Dott. JP. Zendri
10%

Primo Ricercatore at INFN Padova

Local Team Leader for ET INFN-Padova

- › Supervisor of PhD students, M. Toffano and S. Marchetti

Connection with

- › ET squeezing team
- › Quantum Frontiers (Dipartimento di Eccellenza)

Prof. M. Bazzan
10%

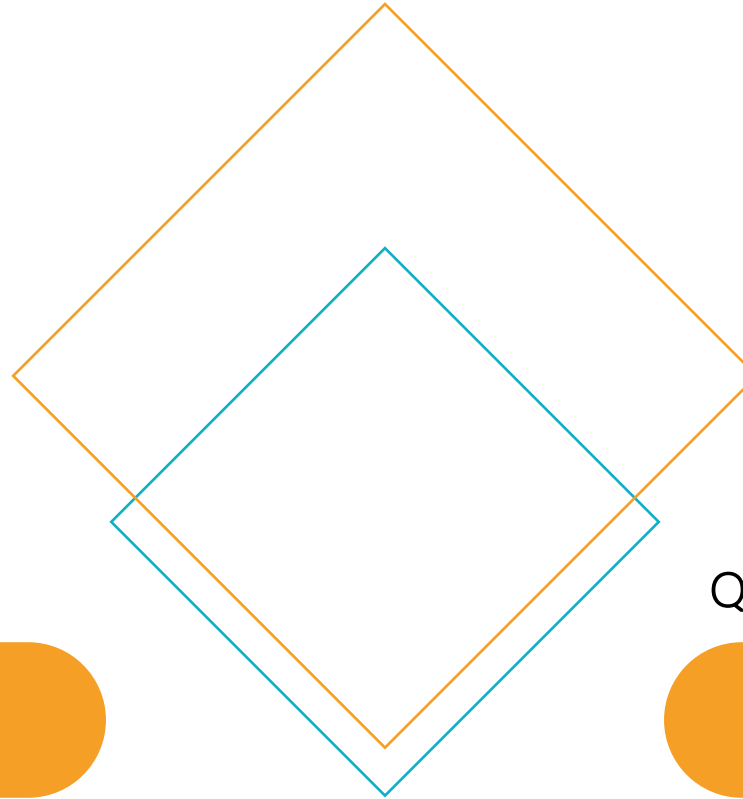
Associated Professor at the
Università degli Studi di Padova

Local Team Leader for SQUEEZE

- › Supervisor of PhD student S. Marchetti
- › Connection with SQUEEZE

WHO?

Official Team



Low noise Electronics

Marco Toffano
20%

Study of Methods for
Low Phase Noise Timing
Distribution in Astrophysics
Experiments

Supervisor: M.Bellato, JP. Zendri

Quantum Optic

Simone Marchetti
20%

Hybrid quantum squeezer

Supervisor: M.Bazzan, JP. Zendri

WHO?

Official Team

Recent Activities

PhD at the University of Trento
Mode Matching Sensing for Quantum
Noise Reduction system in Advanced
Virgo+

PostDoc at Niels Bohr Institute
Atomic Noise for
Gravitational Wave Detectors

Marie Skłodowska-Curie Actions
at Niels Bohr Institute
Einstein-Podolsky-Rosen proxy for
Quantum Memories

Me
A. Grimaldi

Relevant Expertise

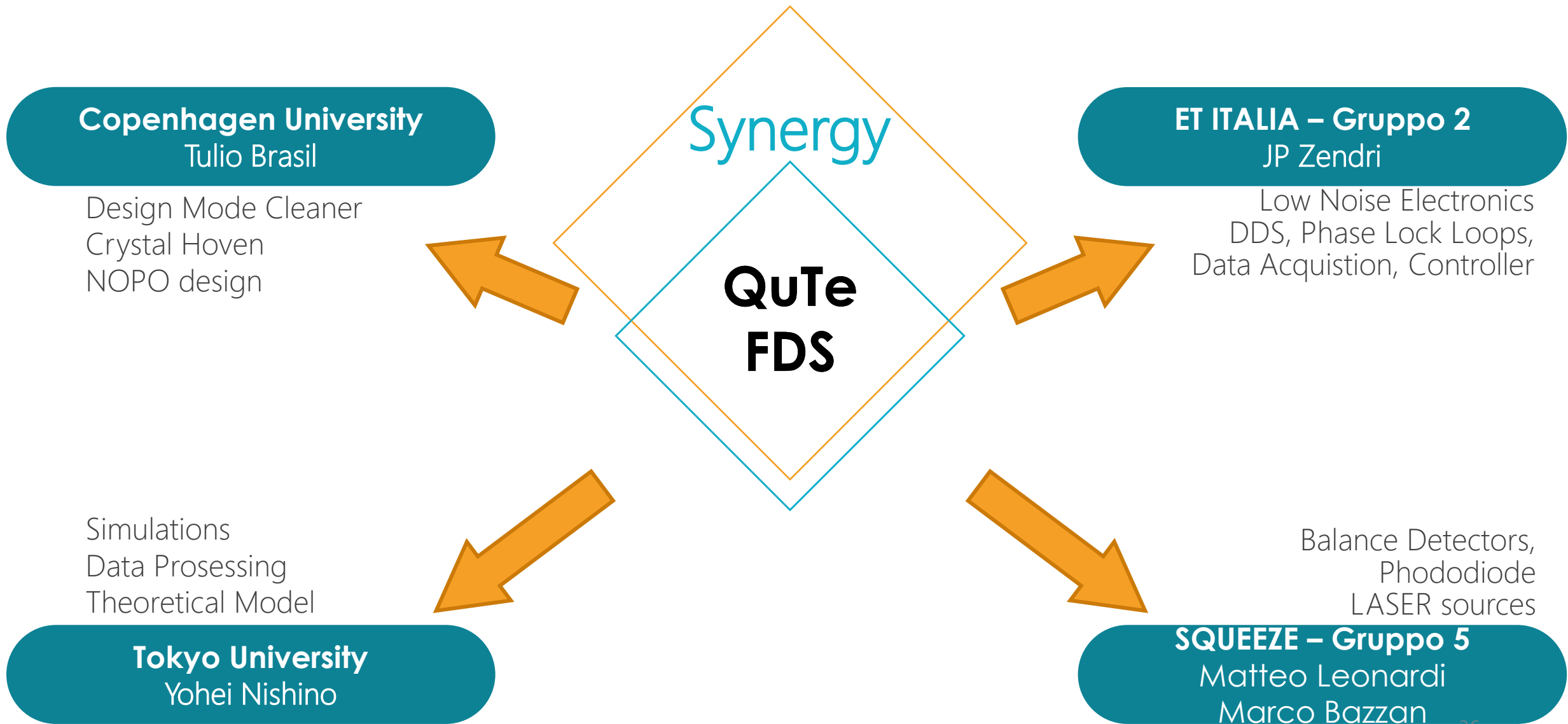
Quantum Noise Reduction technique
for Gravitational Wave Detectors

Two colors EPR squeezing source in
audio bandwidth

Teleportation protocols for continuous
variable

WHO?

External Collaboration



Why

Summary and outlook

	SCIENTIFIC	STRATEGIC
AT THE END	Quantum Noise Reduction › Validation of a new Quantum Noise Reduction Technique. › First investigation about technical limitation of the protocol.	Test bench for other QNR › EPR squeezing source at 1550nm › Single mode squeezer at 1500nm
AFTER	Future experiments › Alternative technique for Quantum Noise reduction in a Gravitational Wave Detector › Multipartite entanglement in continuous variable	Possible external collaboration › Optical & Quantum communications at the University of Padua. Teleportation of non-classical state through telecom fibers › SQUEEZE. In-chip multimode squeezing source